

Assignment 04: Generalization and Regularization Techniques for Neural Networks

CSci 560: Neural Networks and Deep Learning

Description

Welcome to our next assignment on regularization techniques to improve generalization performance in neural networks. Deep Learning models have so much flexibility and capacity that **overfitting can be a serious problem**, if the training dataset is not big enough. Sure it does well on the training set, but the learned network **doesn't generalize to new examples** that it has never seen!

We will be continuing from the last assignment and looking at how we can improve the generalization performance of a neural network model by using regularization. Also we will have a bit of practice overriding the Keras `Callback` classe and using Keras callbacks in this assignment.

Instructions:

- Do not use loops (for/while) in your code, unless the instructions explicitly ask you to do so.
- Use the Keras functional API to implement your asked for network architectures, and to practice and get used to specifying models using that API.
- The workflow has changed a bit in this assignment. You have not been given the function definitions for the functions you will write in the `assg_tasks.py` file. Make sure you correctly declare the function, and always provide PyDoc function documentation for your function. See [Python Docstrings](#) for details on using them and best practices. You are required to use the **NumpyDoc style Docstrings**, where you document the input Parameters and the Return values to each function.
- Also because of this change, you will need to uncomment calls to import your function once you create it and uncomment the code to call the tests on your function. Your functions need to be named exactly as required for the task so that the tests given for functions do not need to be modified to test your code.
- Also see [Pep8 Python Style Guide](#) and follow best practices for formatting your Python code.

Objectives

You will learn:

- to use regularization in your deep learning models.
- the effect of different regularizaiton techniques on performance.
- what weight and Dropout regularization are and how you use them in your Keras models
- How to use and create Keras callbacks by subclassing the `Callback` object from Keras

Overview and Setup

Assignment Tasks

Assignment Submission

Additional Information